

Cloud Computing and Virtualization

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Lecture – 01

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In this lecture...

- I. Introduction to Cloud Computing
 - i. Need of Cloud Computing
 - ii. NIST definition
 - iii. Characteristics and Benefits
 - iv. Applications

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Cloud Computing

- The **cloud** refers to a network of remote servers hosted on the Internet that store, manage, and process data, instead of using a local computer or personal server.
- Google Drive, Dropbox, and AWS S3
- Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet (cloud) without direct active management by the user.

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Need of Cloud Computing

Cost
Efficiency

Scalability and
Flexibility

Reliability and
Availability

Accessibility
and Mobility

Automatic
Updates and
Maintenances

Collaboration
and
Integration

Security

Innovation
and Speed

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NIST Definition of Cloud Computing

[National Institute of Standards and Technology \(NIST\)](#)

“A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

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NIST's Five Essential Characteristics of Cloud Computing

Scalability

- Infrastructure easily scales up or down to meet fluctuating business demands

Lower Costs

- Pay-as-you-go models help control IT costs by charging only for used resources and eliminate equipment purchases

Multiple Storage Options

- Choose from public, private, or hybrid cloud storage based on specific requirements and security needs

Enhanced Data Security

- Advanced security features including granular permissions, authentication, encryption, and virtual private clouds

Flexible Control Options

- Various “as-a-service” models (SaaS, IaaS, PaaS) provide different levels of organizational control.

Improved Accessibility

- Access applications and data from any location, time, and Internet-connected device.

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Benefits of Cloud Computing

- **Scalability:** Infrastructure easily scales up or down to meet fluctuating business demands without hardware investment
- **Lower Costs:** Pay-as-you-go models - charging only for used resources
- **Multiple Storage Options:** Choose from public, private, or hybrid cloud storage based on specific requirements and security needs
- **Enhanced Data Security:** Advanced security features including granular permissions, authentication, encryption, and virtual private clouds
- **Flexible Control Options:** Various “as-a-service” models (SaaS, IaaS, PaaS) provide different levels of organizational control
- **Improved Accessibility:** Access applications and data from any location, time, and Internet-connected device

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Application areas of Cloud Computing

- Data Storage and Backup
- Web Hosting and Application Hosting
- Software as a Service (SaaS)
- Big Data Analytics
- Cloud-Based Development and Testing
- Education and E-Learning
- Healthcare
- Business and Enterprise Applications
- Entertainment and Media Streaming
- Internet of Things (IoT)

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Application areas of Cloud Computing

Task

Write 5 application area of cloud computing with 2 use cases for each field.

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Lecture – 02

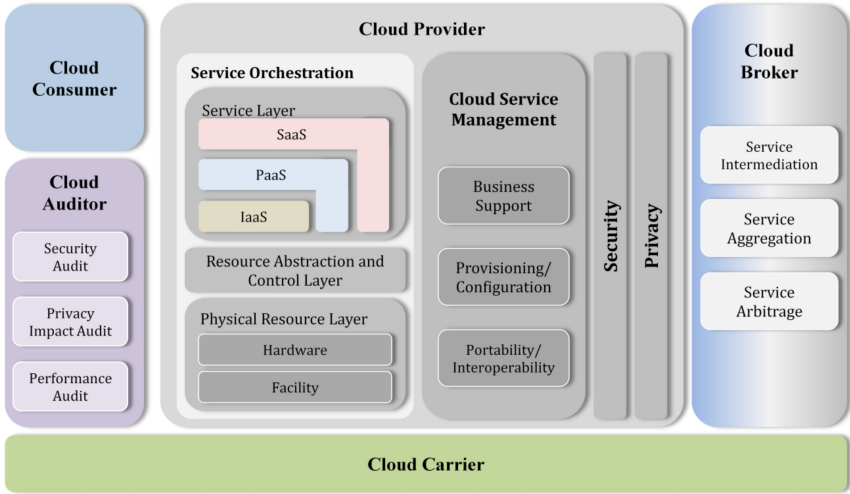
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In this lecture...

- I. Cloud reference model
 - i. NIST architecture
 - ii. Design principles of cloud architecture
 - iii. Infrastructural Constraints
- II. Evolution of Cloud Computing
- III. Cloud service model
 - i. Software as a Service (SaaS)
 - ii. Platform as a Service (PaaS)
 - iii. Infrastructure as a Service (IaaS)

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NIST Reference Architecture



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NIST Reference Architecture

NIST cloud computing reference architecture defines five major actors:

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, <i>Cloud Providers</i> .
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between <i>Cloud Providers</i> and <i>Cloud Consumers</i> .
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from <i>Cloud Providers</i> to <i>Cloud Consumers</i> .

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Design Principles of Cloud Architecture

Scalable

- The ability to increase or decrease resources based on demand.

Flexible

- The ability to adapt quickly to changing requirements and incorporate new technologies.

Highly Available & Resilient

- The ability to remain operational despite infrastructure failures, with minimal or no downtime.

Secure

- Protection against unauthorized access, data breaches, and other security threats through multiple layers of security controls.

Cost Effective

- Optimizing resource usage to minimize unnecessary expenses while meeting performance requirements.

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Design Principles of Cloud Architecture

Task:

For each cloud architecture design principle, identify real-world scenarios where failing to follow that principle negatively impacts system performance, availability, or cost. For every scenario, explain:

- What goes wrong when the principle is not applied.
- How applying the design principle solves the issue, using practical or real-world examples.

Example (Scalability):

Scenario: When Netflix releases a popular show, millions of users log in simultaneously, causing massive traffic spikes.

Problem: Without scalability, fixed servers would become overloaded, leading to buffering, downtime, or service outages.

Solution (Design Principle – Scalability & Elasticity):

Netflix uses AWS Auto Scaling to automatically add thousands of servers during peak demand and reduce them when load decreases. This ensures smooth streaming performance without maintaining expensive servers all year.

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Design Principles of Cloud–Native Architecture

Design for
automation

Be smart with
state

Favor Managed
Services

Practice defense
in depth

Always be
architecting

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Infrastructural Constraints in Cloud Computing

The technical and physical limitations that affect the efficiency, scalability, and reliability of cloud-based systems.

These constraints arise from the underlying hardware, network, storage, and management Infrastructure that supports the cloud environment.

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Infrastructural Constraints in Cloud Computing

Network Bandwidth and Latency

- Limited or unstable network bandwidth can cause delays in data transmission between clients and cloud servers.
- Applications with real-time requirements (e.g., gaming, conferencing) may experience performance degradation.
- Uploading large datasets to cloud storage can take hours with poor internet connectivity.

Power and Cooling Limitations

- Data centers consume massive amounts of electricity and require efficient cooling systems.
- Power shortages or inefficient cooling can lead to downtime or increased operational costs.
- High energy costs in hot regions may limit expansion of data centers.

Storage Scalability and I/O Bottlenecks

- Storage systems may face input/output (I/O) bottlenecks as data volumes grow.
- Reduced performance for data-intensive tasks like big data analytics or AI model training.
- Delays in reading/writing data to disk in shared cloud storage environments.

Data Center Location and Availability

- The physical distance between users and cloud data centers affects latency and service availability.
- Applications with real-time requirements (e.g., gaming, video conferencing) may experience performance degradation.
- Uploading large datasets to cloud storage can take hours with poor internet connectivity.

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Infrastructural Constraints in Cloud Computing

Virtualization Overhead

- Hypervisors and containerization introduce performance overhead compared to native hardware.
- Reduced computational efficiency for high-performance applications.
- Running GPU-intensive workloads in a virtualized cloud may not match bare-metal performance.

Security and Isolation Limitations

- Multi-tenant cloud environments may experience security or privacy risks due to shared infrastructure.
- Vulnerabilities in hypervisors or misconfigurations can lead to data breaches.
- “Side-channel attacks” on co-located virtual machines.

Interoperability and Vendor Lock-in

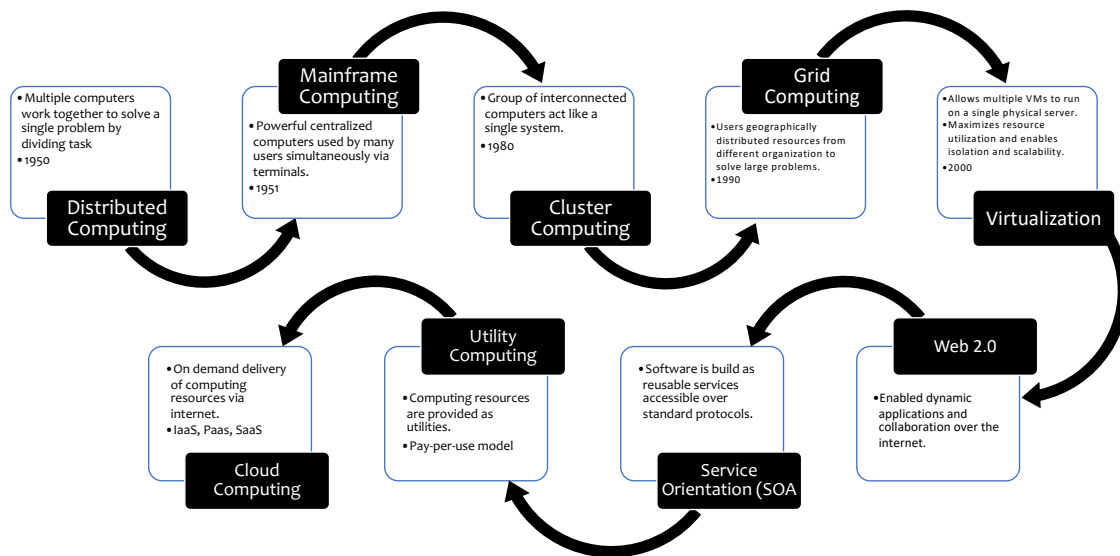
- Different cloud providers use distinct APIs, architectures, and data formats.
- Migrating workloads between clouds can be complex and costly.
- Applications built for AWS may not easily run on Azure without modifications.

Reliability and Downtime Risks

- Even major cloud providers experience service disruptions due to hardware failures or software bugs.
- Temporary unavailability of mission-critical services.
- Cloudflare outage affecting global websites.

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History of Cloud Computing



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Cloud Service Models – IaaS

IaaS – Infrastructure as a Service

- Provides virtualized computing resources over the internet: servers, storage, networks, and operating systems.
- Examples: AWS EC2, Google Compute Engine, Azure Virtual Machines, DigitalOcean Droplets
- Use Cases
 - Hosting custom websites
 - Running virtual machines
 - Building your own databases
 - Big data processing (Hadoop clusters)
 - Creating test and development environments

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Cloud Service Models – PaaS

PaaS – Platform as a Service

- Provides a preconfigured platform (OS, runtime, middleware, development tools) for building and deploying applications.
- Examples: Google App Engine, AWS Elastic Beanstalk, Azure App Service
- Use Cases
 - Rapid application development
 - Deploying APIs
 - Microservices
 - Web apps without server maintenance

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Cloud Service Models – SaaS

SaaS – Software as a Service

- Provides fully managed applications over the internet. Users simply log in and start using the software.
- Examples: Gmail, Google Drive, Microsoft 365, Zoom
- Use Cases
 - Productivity tools
 - CRM systems
 - Accounting software
 - Video conferencing
 - Learning management systems

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Design Principles of Cloud Architecture

Task:

For each cloud service model (IaaS, PaaS, SaaS), identify real-world scenarios where that particular model is the most appropriate choice. For every scenario, briefly explain:

- The situation or problem
- Why the chosen cloud model fits the scenario
- How the model simplifies or improves the solution

Example (PaaS):

A developer wants to deploy a Node.js application without managing servers or configuring infrastructure. Using Google App Engine, they simply upload their code, and the platform automatically handles OS setup, scaling, patching, and load balancing—allowing the developer to focus solely on writing and updating the application.

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Lecture – 03

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In this lecture...

- I. Cloud Deployment model
 - i. Public
 - ii. Private
 - iii. Hybrid
 - iv. Community
- II. Jericho Cloud Cube model
- III. Challenges and ethical issues

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Cloud Deployment Models – Public Cloud

- The cloud service provider owns and manages the entire underlying infrastructure and makes it available for public use.
- All hardware resources and networking components are fully controlled and operated by the service provider.
- The provider is responsible for maintaining the data center, including physical security, hardware upkeep, power, cooling, and overall infrastructure management.
- The provider delivers services through shared infrastructure, ensuring that multiple customers (tenants) can use the same resources while keeping their data strictly isolated and protected with multiple layers of security mechanisms.
- Examples: Microsoft Azure, Amazon AWS, Google Cloud, etc.

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Cloud Deployment Models – Public Cloud

Advantages	Disadvantages
Distributed data centers enhance performance and support users across different regions.	Shared infrastructure may increase exposure to security risks, requiring strong compliance and data protection controls.
Pay-as-you-go pricing eliminates large upfront investments and aligns costs with actual consumption.	Organizations have less visibility and fewer customization options compared to private cloud or on-premise setups.
The cloud provider manages infrastructure maintenance, updates, and system reliability.	Usage-based billing can lead to unexpectedly high expenses if consumption is not closely monitored.
New applications or environments can be provisioned within minutes.	
Resources can be increased or decreased instantly to meet changing workload demands.	
Well-suited for variable workloads, development/testing environments, and disaster recovery scenarios.	

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Cloud Deployment Models – Private Cloud

- The cloud infrastructure is exclusively owned, controlled, and operated by a single organization for its internal use.
- All hardware, storage, networking devices, and data center components are dedicated to one entity, ensuring complete control over computing resources.
- The organization itself (or a third-party vendor hired on contract) manages the infrastructure, including physical security, hardware lifecycle, network configuration, and policy enforcement.
- The entire environment is isolated from external users, offering high levels of customization, governance, and security aligned with organizational requirements.
- The private setup enables organizations to tailor resource allocation, compliance controls, and data protection methods according to their own standards and regulations.

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Cloud Deployment Models – Private Cloud

Advantages	Disadvantages
Full control over infrastructure, configurations, and security policies.	Very expensive to set up and maintain.
Higher level of privacy, compliance, and governance.	Requires skilled IT staff for management.
Stable performance since resources are dedicated.	Limited scalability compared to public cloud.
Customizable environment based on organizational need.	Hardware upgrades and capacity planning remain the organization's responsibility.

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Cloud Deployment Models – Hybrid Cloud

- The hybrid cloud integrates two or more cloud environments typically a combination of public and private clouds that operate as a unified system.
- Organizations can run sensitive or regulated workloads in the private cloud while shifting less-critical or highly elastic workloads to the public cloud.
- Data and applications move securely between environments through encrypted connections, VPNs, APIs, or cloud interconnects.
- The infrastructure provides flexibility by allowing organizations to scale resources dynamically using public cloud capacity while retaining internal control of critical systems.
- Proper orchestration and management tools ensure consistent policy enforcement, workload distribution, and monitoring across both environments.

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Cloud Deployment Models – Hybrid Cloud

Advantages	Disadvantages
High flexibility: use private cloud for sensitive workloads and public cloud for scalable tasks.	Complex architecture requiring strong integration and management tools.
Better cost optimization compared to fully private cloud.	Potential security risks if data flows are not configured properly.
Improved disaster recovery and backup options.	Requires expertise to maintain interoperability.
Seamless scalability using public cloud resources.	Higher initial configuration cost than pure public cloud

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Cloud Deployment Models – Community Cloud

- The cloud infrastructure is shared among several organizations that have common requirements such as regulatory needs, operational goals, or security concerns.
- All participating organizations jointly own or subscribe to the shared infrastructure, reducing costs while ensuring that specialized needs are met.
- The cloud may be managed by one of the participating organizations or by a third-party provider dedicated to serving the community.
- It provides controlled access where only authorized community members can use the shared resources, ensuring heightened security and compliance for the specific domain or group.
- The setup supports collaboration by offering a common platform for shared projects, workloads, or data exchange under mutual policies.
- Examples: Government community clouds, Research or education community clouds.

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Cloud Deployment Models – Community Cloud

Advantages	Disadvantages
Cost-effective model for organizations with similar requirements.	Higher cost than public cloud due to limited user base.
Higher security and compliance than public cloud.	Not as customizable as private cloud.
Shared infrastructure reduces duplication of effort.	Governance can be complicated due to shared ownership.
Encourages inter-organization collaboration.	Requires trust and coordination between participating organizations.

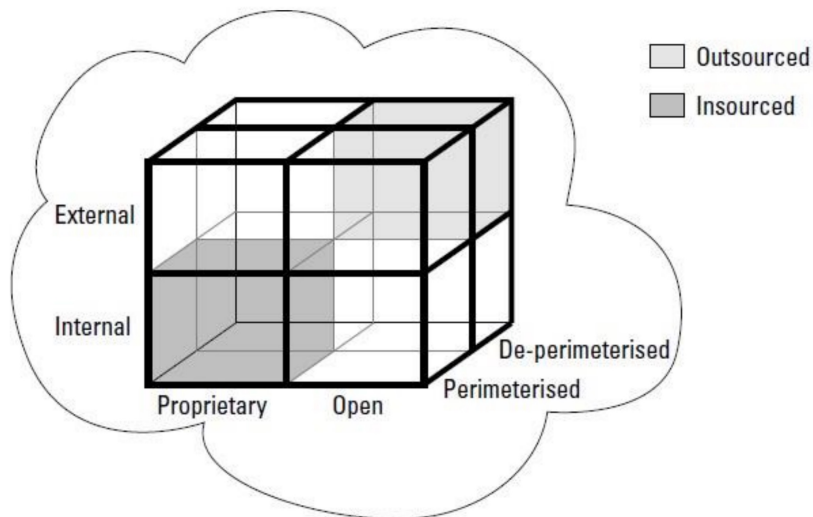
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Jericho Cloud Cube Model

- The Jericho Cloud Cube Model is a framework created by the Jericho Forum to help organizations understand different cloud deployment scenarios and assess the security implications of each.
- This model uses four dimensions to classify cloud environments based on how data, services, and infrastructure are deployed and secured.
- The model is a 3D cube with four key dimensions:

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Jericho Cloud Cube Model



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Jericho Cloud Cube Model

Internal vs. External

- This dimension describes where the cloud infrastructure physically resides.

Internal

- Infrastructure is located within the organization's own data center.
- Example: Private cloud built on-premise.

External

- Infrastructure is owned/hosted by a third-party provider.
- Example: AWS, Azure, Google Cloud, IBM Cloud.

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Jericho Cloud Cube Model

Proprietary vs. Open

- This dimension addresses who controls and owns the technology.

Proprietary

- Closed systems, vendor-specific technologies.
- Might result in vendor lock-in.
- Example: AWS proprietary tools (Lambda, DynamoDB, etc.)

Open

- Open standards, open-source technologies.
- Easier interoperability and migration.
- Example: OpenStack-based clouds.

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Jericho Cloud Cube Model

Perimeterised vs. De-perimeterised

- This dimension focuses on security boundaries.

Perimeterised

- Traditional security model.
- Trust and protection based on network boundaries (firewalls, VPNs).
- Example: Private cloud within corporate LAN.

De-perimeterised

- Security does NOT rely on perimeter defense.
- Zero-trust style approach.
- Identity and encryption are key.
- Typical in public and hybrid cloud models.

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Jericho Cloud Cube Model

Insourced vs. Outsourced

- This is often shown as a color band on the diagram.

Insourced

- Managed by the organization's own IT team.

Outsourced

- Managed entirely by a third-party provider or vendor.

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Challenges in Cloud Computing

Security and Privacy Risks

- Risks include data breaches, insecure APIs, and unauthorized access.
- Maintaining strong encryption, IAM, and compliance controls is essential.

Data Governance and Compliance

- Different countries have different data protection laws.
- Ensuring compliance in cross-border cloud environments is difficult.

Vendor Lock-In

- Applications and data become dependent on one provider's proprietary tools or formats.
- Moving to another provider is costly and technically challenging.

Downtime and Service Reliability

- An outage can disrupt mission-critical services and cause revenue loss.
- Dependence on internet connectivity increases vulnerability.

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Challenges in Cloud Computing

Limited Control over Infrastructure

- Troubleshooting and performance tuning can be difficult.
- Customization is limited compared to on-premise environments.

Performance and Latency Issue

- Services hosted in distant data centers can introduce network delays.
- High-volume or real-time applications (IoT, trading systems) may suffer.

Cost Management Challenges

- Pay-as-you-go billing can lead to unexpected or hidden costs.
- Without monitoring, cloud usage can rapidly exceed budgets.

Integration and Migration Complexity

- Moving legacy systems or data into the cloud can be expensive and time-consuming.
- Hybrid cloud integration requires advanced networking and orchestration.

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Ethical Issues in Cloud Computing

Data Privacy and User Consent

- Data is shared without explicit consent.
- Users are unaware of how data is used.
- Data is mined for analytics or advertising.

Data Ownership and Control

- Who truly owns data stored in the cloud?

Transparency Issue

- Lack of transparency undermines user trust.

Surveillance and Governance

- Mass surveillance, privacy invasion, misuse of personal data.

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Ethical Issues in Cloud Computing

Environmental Impact

- Carbon footprint
- Non-renewable energy use
- E-waste and hardware disposal

Digital Divide

- Creates ethical concerns about fairness, inclusiveness, and equal opportunity.

Fair Billing and Transparency

- Users may be billed for resources they didn't intend to consume.

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Ethical Issues in Cloud Computing

Task

Find ethical issues in cloud computing and explain them with:

- Ethical issues
- Stakeholders
- Risks
- Solutions

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References

- NIST Reference Architecture:
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- Design Principle of Cloud Native Architecture:
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